Project 3 Report

INFSCI 2591 - Algorithm Design

Abhay Kumara Sri Krishna Nandiraju
University of Pittsburgh

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1 O1. Theoretical Time Complexity Analysis

1.1 A. Time Complexity of Nearest Neighbor Algorithm

Analysis:

- Let N be the number of nodes in the graph
- The outer loop in the algorithm runs for N-1 times to make sure that every node is visited.
- The inner loop searches all the N nodes to find the minimum edge weight from the current node to any unvisited node.
- Since, outer loop runs $N-1$ times and inner loop runs N times, the overall time complexity of the Nearest Neighbor algorithm is $O(N(N-1)) = O(N^2)$.

1.2 B. Time Complexity of Nearest Insertion Algorithm

Analysis:

- It takes $O(N^2)$ to find the shortest initial edge since it requires checking all pairs of nodes.
- The outer while loop runs for N-2 times and inside the loop, finding the closest unvisited node requires comparing all remaining unvisited nodes against all nodes. In the worst case, this is $O(N^2)$.
- Evaluating the insertion cost at every possible best position in the tour takes $O(N)$ time.
- Overall time complexity is $O((N-2)(N^2+N)) = O(N^3 - 2N^2 + N^2 + 2N) = O(N^3)$

2 O2. Results Table

Algorithm	File Name	Tour: Sequence of Nodes	Total Weight
Nearest Neighbor	4n.csv	0,1,2,3,0	2611.0
	5n.csv	0,3,2,4,1,0	1290.0
	6n.csv	0,5,3,4,1,2,0	3198.0
	7n.csv	0,6,4,3,1,5,2,0	2425.0
	8n.csv	0,2,4,1,6,3,5,7,0	2898.0
	9n.csv	0,8,1,6,5,2,3,7,4,0	2842.0
	10n.csv	0,7,8,4,2,6,5,3,1,9,0	3008.0
	11n.csv	0,1,6,4,8,2,9,7,5,10,3,0	3350.0
	12n.csv	0,8,3,5,6,4,7,11,10,2,1,9,0	2646.0
	13n.csv	0,9,2,11,7,12,3,6,1,5,4,8,10,0	3653.0
Nearest Insertion	4n.csv	0,3,2,1,0	2611.0
	5n.csv	0,3,2,4,1,0	1290.0
	6n.csv	0,5,3,2,4,1,0	2455.0
	7n.csv	0,1,5,2,3,4,6,0	1993.0
	8n.csv	0,2,4,5,3,7,1,6,0	2692.0
	9n.csv	0,8,4,2,5,6,1,7,3,0	2588.0
	10n.csv	0,9,1,3,8,4,5,6,2,7,0	3423.0
	11n.csv	0,3,5,2,9,7,8,4,10,6,1,0	2867.0
	12n.csv	0,4,7,11,10,6,5,1,2,3,8,9,0	2930.0
	13n.csv	0,11,8,3,6,4,5,1,10,7,12,2,9,0	3031.0
Dynamic Programming	4n.csv	0,1,2,3,0	2611.0
	5n.csv	0,1,2,4,3,0	1146.0
	6n.csv	0,1,4,2,3,5,0	2455.0
	7n.csv	0,1,5,2,3,4,6,0	1993.0
	8n.csv	0,2,4,1,7,5,3,6,0	2667.0
	9n.csv	0,3,2,4,1,6,5,7,8,0	2439.0
	10n.csv	0,5,6,2,4,9,1,3,8,7,0	2953.0
	11n.csv	0,1,6,10,5,2,9,7,8,4,3,0	2701.0
	12n.csv	0,6,4,7,5,3,8,1,2,10,11,9,0	1907.0
	13n.csv	0,11,2,9,10,7,8,4,5,1,6,3,12,0	2523.0
Branch and Bound	4n.csv	0,1,2,3,0	2611.0
	5n.csv	0,1,2,4,3,0	1146.0
	6n.csv	0,1,4,2,3,5,0	2455.0
	7n.csv	0,1,5,2,3,4,6,0	1993.0
	8n.csv	0,2,4,1,7,5,3,6,0	2667.0
	9n.csv	0,3,2,4,1,6,5,7,8,0	2439.0
	10n.csv	0,5,6,2,4,9,1,3,8,7,0	2953.0
	11n.csv	0,1,6,10,5,2,9,7,8,4,3,0	2701.0
	12n.csv	0,6,4,7,5,3,8,1,2,10,11,9,0	1907.0
	13n.csv	0,11,2,9,10,7,8,4,5,1,6,3,12,0	2523.0

Table 1: Test Case Results for all 4 algorithms